

vSphere

Use **vSphere** as a VI administrator to virtualize physical infrastructure into managed CPU, storage, and networking resources.

The two core components of VMware vSphere are ESX and vCenter. ESX is the virtualization platform where you create and run virtual machines and virtual appliances, Kubernetes workloads, and AI/ML workloads. vCenter is the service through which you manage multiple hosts connected in a network and pool host resources.

vSphere manages large collections of infrastructure, such as CPUs, storage, networking, GPUs, and DPUs as a seamless and dynamic operating environment, and also manages the complexity of a data center.

vSphere Functional Area	Capabilities
Lifecycle management of vSphere	• Lifecycle management via pre-staging ESX images, remediating hosts in parallel, and by applying updates in parallel across clusters. • vSphere can manage infrastructure images to patch, update, or upgrade clusters using a desired state model. • vSphere can reduce maintenance windows via reduced downtime upgrades for vCenter as well as live patching for ESX for near zero downtime for security patches.
Workload performance	• With its GPU support, vSphere improves workload performance for modern AI workloads • vSphere on DPUs (VMware vSphere® Distributed Services Engine™) helps accelerate infrastructure network functions on the Data Processing Unit (DPU) by utilizing available CPU cycles achieve higher workload consolidation per host. • Distributed Resource Scheduler (vSphere DRS) enables automatic load balancing of resources allocated to workloads in a vSphere cluster. Storage DRS optimizes VM data placement as the VM is created and used over time.
Business continuity	• High Availability with vSphere automatically restarts your VMs following physical machine failure. • Fault Tolerance provides continuous availability of any application in the event of a hardware failure with no data loss or downtime. • vMotion enables live migration of virtual machines with no disruption to users or loss of service, eliminating the need to schedule application downtime for planned server maintenance. Storage vMotion avoids downtime for planned storage maintenance.
Built-in security	• Identity federation with Microsoft Entra ID (formerly Azure AD), ADFS and Okta: Secure access and account management. • Virtual Machine Encryption: Data-at-rest encryption for virtual machine data and disks. • vSphere Trust Authority: Remote attestation for sensitive workloads.

vSphere Documentation

For information about the built-in vSphere capabilities, see the [vSphere documentation](#).

Using the VCF Capabilities for vSphere

VCF fully automates vCenter deployment and management in all infrastructure management including deploying VCF, licensing, and adding workload domains, vSphere clusters and hosts to your environment. See the following documentation for information on these management and operational capabilities..

Learn More About	Documentation
Deploying a new VCF platform or using existing vSphere infrastructure to deploy a new VCF platform or VCF instance	Deployment, Convergence, and Upgrade
Adding workload domains, clusters and ESX hosts	Building Cloud Infrastructure
Managing licenses	Licensing
Integrating cloud infrastructure in VCF Automation if you are a provider administrator	Provider Management
Performing operational and monitoring tasks on cloud infrastructure	• Fleet Management • Infrastructure Operations

Lifecycle Management

Use **VCF Operations** as a VI administrator to manage the lifecycle of the management and SDDC components in VCF, including downloading binaries and updating VCF fleet and VCF Instances.

For VCF, you use VCF Operations for these capabilities. For vSphere Foundation, you perform the supported operations on the individual management components.

In This Chapter

Lifecycle Management of VCF Management Components

- [Lifecycle Management in VMware Cloud Foundation](#)
- [Lifecycle Management of VCF Core Components](#)
- [Lifecycle Management of VCF Management Components](#)
- [Using the VCF Download Tool](#)

Getting to Know VCF

To apply this documentation, you must be acquainted with the [VCF Product Overview](#), [VCF Design](#), and [VCF Release Notes](#).

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See [Administration SDKs, APIs, and CLI](#).

Lifecycle Management in VMware Cloud Foundation

VCF Operations provides a comprehensive set of tools for managing the lifecycle of both VCF management components and VCF core components. You can use VCF Operations to download and manage upgrade, patch, and install binaries. You can also manage ESX components and vSphere Lifecycle Manager images and deploy optional VCF management components.

VCF Component Classification

VCF components are classified based on the level at which they operate. VCF Management components operate on the fleet level, while VCF core components operate on the instance level.

VCF Management Components	VCF Core Components
• VCF Operations • VCF Automation • VCF Operations fleet management • VCF Operations for logs • VCF Operations for networks • VCF Identity Broker	• SDDC Manager • vCenter • NSX Manager • ESX

VCF Release Types

The VCF release policy defines four distinct release types, each with a specific purpose, scope, and set of lifecycle rules.

Type	Description
Major	A major release, such as 9.0.0.0 or 10.0.0.0, represents the most significant evolution of the VCF platform. It serves as the primary vehicle for introducing major features and changes that can redefine the platform's capabilities. This is the only release type that can introduce breaking changes or remove existing functionality. However, this is governed by a strict deprecation policy designed to protect enterprise stability: a feature must be officially announced for deprecation in a prior major release before it can be removed in a subsequent one. Major releases are fully synchronized, meaning all VCF components are updated in lockstep as part of a single release. They follow a unified consumption pattern, requiring a full-stack upgrade.
Minor	A minor release, such as 9.1.0.0 or 9.2.0.0, is the primary mechanism for delivering new features and enhancements within a major version. These releases include features and are designed to be functionally additive. Minor releases aim to be non-disruptive from a compatibility and API perspective. Like major releases, minor releases are fully synchronized and follow the unified consumption pattern. The policy that "All components are updated in a major and minor release" applies here, ensuring the entire platform advances to a new, consistent feature level.
Maintenance	A maintenance release, such as 9.0.1.0 or 9.0.2.0, is a scheduled, cumulative update, focused on enhancing platform stability and support. Maintenance releases provide bug and security fixes, hardware enablement changes, such as support for servers, CPUs, devices, driver updates, guest OS support, and backward-compatible new features. They are limited in scope to minimize the operational impact. You can pick and choose which components to apply based on the impact on your environment and operational costs. For example, if you run into a critical issue with vCenter, you do not need to apply a patch to ESX that is irrelevant to your environment. Maintenance releases are synchronized among VCF components with a new Bill Of Materials.
Patch	Patch releases, such as 9.0.1.0100, provide time-sensitive fixes to security and catastrophic issues, impacting business operations, that need to be released between major, minor, and maintenance releases. You can pick and choose which components to apply based on the impact on your environment and operational costs. Patch releases are not synchronized among all VCF components and do not require a new implementation of a Bill Of Materials.

Lifecycle Management in VCF Operations

With VCF 9.0 and later, you use VCF Operations to manage major, minor, maintenance, and patch releases for your VCF environment.

The VCF management components and VCF core components are managed separately, although you can configure them with the same online or offline depot. The management components are managed at the VCF fleet level and can manage more than one VCF Instance. The core components are managed at the VCF Instance level.

For upgrading your management domain to VCF 9.0, see [Upgrading Your VCF Management Domain to VMware Cloud Foundation 9.0](#).

For upgrading your workload domains to VCF 9.0, see [Upgrading VMware Cloud Foundation 5.x Workload Domains to 9.0](#).

Update Order for Components for Maintenance Releases

When you update your VCF environment to a maintenance release version, such as 9.0.1.0 or 9.0.2.0, you follow a specific order:

1. Update your management components that operate on the fleet level in a predefined order. See [Lifecycle Management of VCF Management Components](#).
 - a. VCF Operations fleet management appliance
 - b. VCF Operations
 - c. Update the remaining components VCF Operations for logs, VCF Automation, VCF Identity Broker, and VCF Operations for networks in your preferred order.
2. Update the core components in the management domain of your VCF Instance in a predefined order. See [Lifecycle Management of VCF Core Components](#).
 - a. SDDC Manager
 - b. NSX
 - c. vCenter
 - d. ESX hosts
 - e. vSAN
3. Update the core components in the workload domains in the same order as the management domain. See [Lifecycle Management of VCF Core Components](#).

For the management components, start with the VCF Operations fleet management appliance, followed by the VCF Operations instance and then proceed with the remaining components in your preferred order. For the core components, start with SDDC Manager, followed by NSX, vCenter, ESX hosts, and vSAN.

Binary Management for VMware Cloud Foundation

Starting with VCF 9, you perform components lifecycle management by using the VCF Operations UI. You download the required binaries for core and management components to the following VCF components: VCF Operations fleet management and SDDC Manager.

- VCF Operations fleet management handles binaries for the VCF management components. See [Binary Management for VCF Management Components](#).
- SDDC Manager handles binaries for the VCF core components for each VCF Instance. See [Binary Management for VCF Core Components](#).

Binary Management for VCF Management Components

VCF operations fleet management handles binaries for the VCF management components.

The way VCF operations fleet management receives required binaries depends on your environment and the download method you configure.

The method that you use to download binaries depends on how you access the Internet in your environment.

- Online depot. The VCF operations fleet management appliance can connect to the Internet, either directly or through a proxy.

- Offline depot - web server. The VCF operations fleet management appliance cannot connect to the Internet, but can connect to a computer with Internet access.
- Offline depot - local path. The VCF operations fleet management appliance cannot connect to the Internet or to a computer with Internet access and you host binaries in a folder within the appliance.

Connect to an Online Depot for VCF Operations fleet management

You connect your VCF Operations fleet management appliance to the online depot which provides a direct source for downloading installation, upgrade, and patch binaries.

Verify that you have valid credentials for your organization to access the [Broadcom Support Portal](#).

With the download token generated from the Broadcom Support Portal, perform the following steps to configure your depot with a unique ID that enables connection to the online depot.

1. Generating a download token for VCF Operations fleet management.
 - a) Log in to the [Broadcom Support Portal](#) and select **VMware Cloud Foundation**.
 - b) On My Dashboard, in the **Quick Links** box that appears, select **Generate Download Token**.
 - c) Find your Site ID on the list displayed and add it as input. Then click **Generate Token**.
 - d) Copy the token to use in VCF Operations fleet management appliance.
2. (Optional) If your environment has no direct Internet access, configure a proxy server.
 - a) In VCF Operations, select **Administration > Global Settings**.
 - b) In the left pane, select **Network Settings**.
 - c) On the **Network settings** page, configure the proxy settings and click **Save**.
3. In VCF Operations, select **Fleet Management > Lifecycle > VCF Management > Depot Configuration**.
4. On the Online Depot tile, click **Configure**.
5. The **Configure Online Depot** form appears.
 - a) Click the plus sign to add a password. Enter the following information, then click **Add**.
 - Password Alias: Enter a name to identify the download token.
 - Password: Enter the download token.
 - Confirm Password: Enter the download token again.
 - Password Description is optional.
 - User Name is optional.
 - b) Click **Select Download Token**.
Scroll to find the Password Alias that you added and select it.
 - c) Click to accept the imported certificate and click **OK**.

When the connection to the online depot is established, the Online Depot tile appears with an active depot connection and the binaries to download appear on the Binary Management tab. See [Using the Binary Management tab](#).

Connect to an Offline Depot for VCF Operations fleet management

Offline depots simplify artifact distribution for air-gapped or dark-site environments, reducing the steps that are needed to manage multiple VCF Instances.

To set up an offline depot either with a web server or a local server, a dedicated virtual machine must be available along with the details of your Bundle Transfer Utility server. For information about the requirements and process for creating an offline depot, see [Set Up an Offline Depot Web Server for VMware Cloud Foundation](#).

Configure an Offline Depot with a Web Server

An offline depot that is a self-managed web server is suited for users operating in an air-gapped network who want to stream binary downloads.

- [Set Up an Offline Depot Web Server for VMware Cloud Foundation](#).
- Verify that you have downloaded binaries for the VCF management components to an internal web server. See [Download VCF Management Component Binaries to an Offline Depot](#).

On offline depot with a web server provides the following advantages:

- Creates a depot-like directory structure.
- Provides bundle management mechanisms.
- Manages upgrade metadata.

Perform the following steps to configure an offline depot with a web server.

1. In VCF Operations, select **Fleet Management > Lifecycle > Depot Configuration**.
2. On the Offline Depot tile, click **Configure**.

To configure the offline depot, enter:

- Offline Depot Type: **Webserver**
- Repository URL: <URL of the web server>
- Username: Enter username with authentication rights on the web server.
- Password: Enter password used to access the web server.
- Click to view and accept the imported certificate.

When the web server connection is established, the offline depot connection appears active. The binaries that you can download from the web server appear on the Binary Management tab. See [Downloading VCF Management Components Binaries](#).

Configure an Offline Depot with a Local Path

An offline depot that is a local server is suited for users operating in a dark site. These infrastructure-secure environments without direct Internet access are typically designed for sensitive or high-security operations.

Verify that you have downloaded binaries for the VCF management components to a local server with Internet access and copied them to a partition of the VCF Operations fleet management appliance such as `/data`. See [Download VCF Management Component Binaries to an Offline Depot](#).

Perform the following steps to configure an offline depot with a local server.

1. In VCF Operations, select **Fleet Management > Lifecycle > Depot Configuration**.
2. On the Offline Depot tile, click **Configure**.

To configure the offline depot, enter:

- Offline Depot Type: **Local Path**
- Path: <Path to the bundle on the VCF Operations fleet management appliance such as `/data`>

Click **OK**.

The offline depot connection appears active and the bundles or binaries that are on your VCF Operations fleet management appliance appear on the Binary Management tab. See [Downloading VCF Management Components Binaries](#).

Downloading VCF Management Components Binaries

Binary management provides a centralized interface that administrators use to download component binaries for upgrades, patches, and installations.

Note:

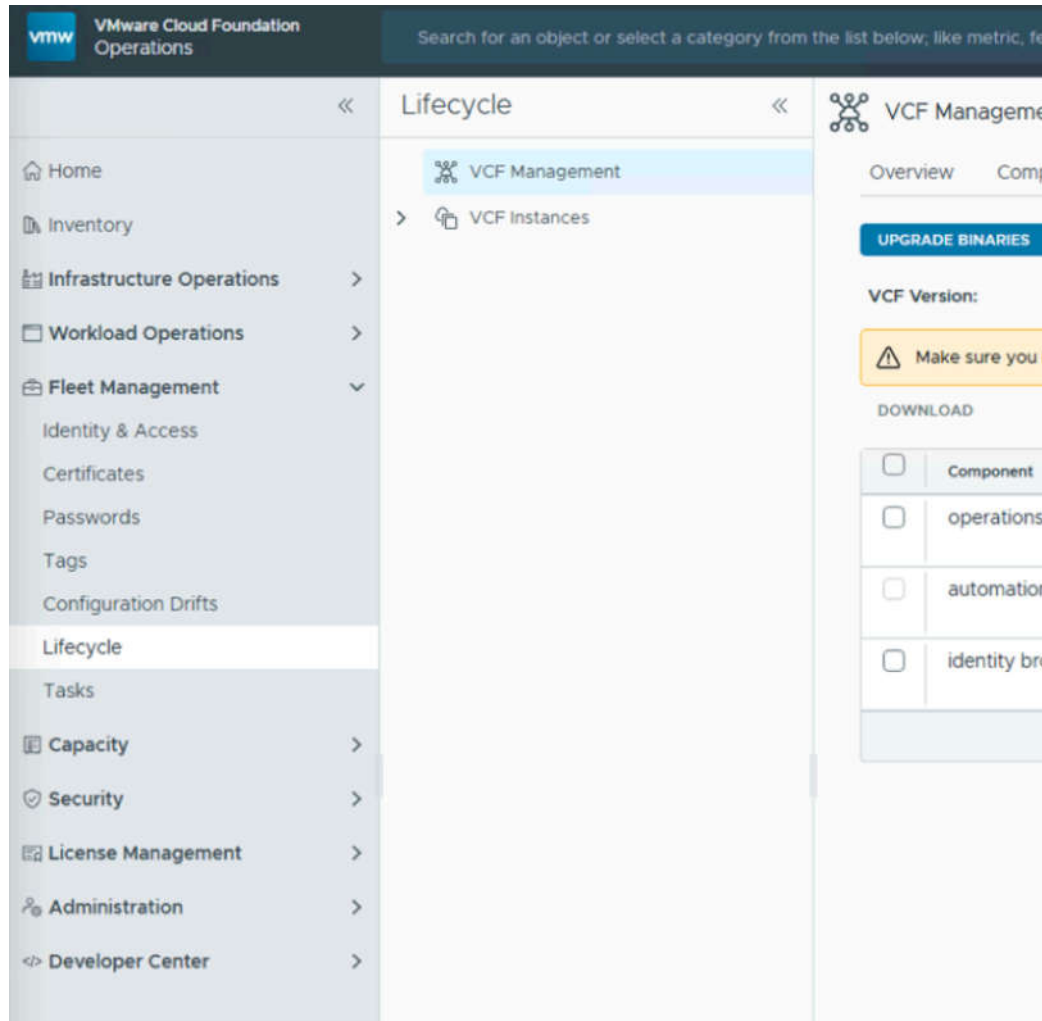
Before you download a binary, you must set up your depot configuration. See [Lifecycle Management of VCF Management Components](#).

To download a binary, select **Fleet Management > Lifecycle > Binary Management** and select the binary type, the VCF version, and the component for the request. Before you click **DOWNLOAD**, ensure that you have selected the correct upgrade, patch, or install binary type. In that way, the correct button appears when you are ready to upgrade, manage, or add your component.

To remove a binary, check the downloaded component and click the trashcan icon.

Figure 156: Binary Management tab

Select the component binary to download or remove.



Download VCF Management Component Binaries to an Offline Depot

Use the VCF Download Tool to download binaries for the VCF management components to an offline depot.

- Generate a download token on the Broadcom Support Portal.
- [Set Up an Offline Depot Web Server for VMware Cloud Foundation](#). (only required if your offline depot is a web server)
- Connect to the offline depot. See [Connect to an Offline Depot for VCF Operations fleet management](#).
Note: You can also connect to the offline depot after you download binaries to the offline depot.
- An offline depot web server must have:
 - The latest version of the VCF Download Tool on the offline depot web server. You can download it from the Broadcom Support portal.
 - Internet connectivity (either directly or through a proxy) for downloading the binaries.
 - Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VCF Download Tool for long-running operations.
- To download binaries when your offline depot cannot connect to the Internet, you must have:
 - A Windows or Linux computer with Internet connectivity (either directly or through a proxy) for downloading the binaries.
 - The latest version of the VCF Download Tool available on the computer with Internet access. You can download it from the Broadcom Support portal.
 - Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VCF Download Tool for long-running operations.

The process varies slightly depending on whether your offline depot is a web server or is using a local path on the VCF Operations fleet management appliance.

If the offline depot can only access the Internet using a proxy server, use the proxy server options as described in [VCF Download Tool Update Manager Download Service \(UMDS\) Commands](#).

Note: VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

1. Download the VCF management component binaries.

Run the command from the offline depot web server or the computer with Internet access, depending on the type of offline depot you have.

```
./vcf-download-tool binaries download --depot-store depot_store --depot-download-token-file token_file_path --vcf-version x.x.x.x --lifecycle-managed-by VRSLCM --type UPGRADE
```

Variable	Replace With
<code>depot_store</code>	For an offline depot web server, enter the offline depot root directory. For a computer with Internet access, enter a path to the directory where the binaries should be downloaded.
<code>token_file_path</code>	The path to a text file containing a single line with your depot download token. You can generate a download token in the Broadcom Support Portal. See step 1 in Connect to an Online Depot for VCF Operations fleet management .
<code>x.x.x.x</code>	The version of VMware Cloud Foundation for which you want to download upgrade/patch binaries. For example: 9.0.0.0.

This command downloads the upgrade/patch binaries for VCF management components. Specify `INSTALL` as the `--type` to download install bundles. For information about other options available in the VCF Download Tool, see [VCF Download Tool Binaries Commands and Options](#).

2. If you downloaded the binaries to a computer with Internet access, copy the binaries directory to the VCF Operations fleet management appliance. If data space is low, use fleet management to extend the partition before copying the binaries. See [Extending storage for binaries](#).

Copy the binaries directory to the path that you specified when you created the offline depot. For example, `/data`.

Extending Storage for Binaries

The `/data` partition is the largest partition in the VCF Operations fleet management appliance and is where you upload binaries. If storage space in `/data` is low, you can extend the partition.

To extend storage space for the installation, upgrade, or patch binaries, you update the system information settings.

1. Select **Fleet Management > Lifecycle > Settings** and click **Extend Storage**.
 - For vCenter Hostname: Enter the vCenter Server FQDN.
 - For vCenter Credential: Select the vCenter Server password.
 - For Disk Size: Enter the desired partition size.
2. Click **Extend**.

Binary Management for VCF Core Components

SDDC Manager handles binaries for the VCF core components.

The way SDDC Manager receives required binaries depends on your environment and the download method you configure.

The method that you use to download binaries depends on how you access the Internet in your environment.

- Online depot. The SDDC Manager appliance can connect to the Internet, either directly or through a proxy.
- Offline depot. The SDDC Manager appliance cannot connect to the Internet, but can connect to a computer with Internet access.
- No depot. The SDDC Manager appliance cannot connect to the Internet or to a computer with Internet access.

Connect to an Online Depot for a VCF Instance

Connect to an online depot for your VCF Instance to access upgrade, patch, and install binaries.

- To connect to an online depot, SDDC Manager must be able to access the Internet, either directly or through a proxy server. See [Configure a Proxy Server for SDDC Manager](#).
- If your environment is using a TLS interception and / or TLS termination proxy with a custom CA certificate to connect to the Internet, you must import the proxy's CA certificate to the SDDC Manager trust store. See [KB 316056](#).
- Generate a download token on the Broadcom Support Portal.
- See [Public URL list for SDDC Manager](#) for information about the URLs that must be accessible to download binaries.

SDDC Manager supports two types of software depots:

- Online depot
- Offline depot

You can only connect SDDC Manager to one type of depot. If SDDC Manager is connected to an online depot and you configure a connection to an offline depot, the online depot connection is disabled and deleted.

If you configured an online depot in VCF Installer, the VCF Instance inherits the depot settings.

1. In VCF Operations, click **Fleet Management > Lifecycle**.
2. Click **VCF Instances** and click the name of a VCF Instance.
3. Click **Depot Settings**.
4. Click **Authenticate** to set up an online depot.
5. Paste the download token that you generated on the Broadcom Support Portal in the text box and click **Authenticate**.
SDDC Manager attempts to connect to the depot. If the connection is successful, SDDC Manager starts looking for available binaries. To view available binaries, click the **Binary Management** tab. It may take some time for all available binaries to appear.

Configure a Proxy Server for SDDC Manager

If SDDC Manager does not have direct Internet access, you can configure a proxy server to access binaries. SDDC Manager supports proxy servers with authentication.

1. In VCF Operations, click **Fleet Management > Lifecycle**.
2. Click **VCF Instances** and click on the name of a VCF Instance.
3. Click **Proxy Settings**.
4. Click **Set Up Proxy**.
5. Toggle the **Proxy Enabled** setting to the on position.
6. Select **HTTP** or **HTTPS**.
7. Enter the proxy server IP address and port number.
8. If your proxy server requires authentication, toggle the **Authentication** setting to the on position and enter the user name and password.

Note:

The supported proxy authentication methods are Basic and NTLMv1.

9. Click **Save**.

Connect to an Offline Depot for a VCF Instance

Connect to an offline depot for your VCF Instance to access upgrade, patch, and install binaries.

To connect to an offline depot, you must first configure it. See [Set Up an Offline Depot Web Server for VMware Cloud Foundation](#).

SDDC Manager supports two types of software depots:

- Online depot
- Offline depot

You can only connect SDDC Manager to one type of depot. If SDDC Manager is connected to an online depot and you configure a connection to an offline depot, the online depot connection is disabled and deleted.

You can connect multiple VCF Instances to the same offline depot.

To download binaries to an offline depot, see [Download VCF Core Component Binaries to an Offline Depot](#).

1. In VCF Operations, click **Fleet Management > Lifecycle**.
2. Click **VCF Instances** and click on the name of a VCF Instance.
3. Click **Depot Settings**.
4. Click **Set Up** to set up an offline depot.
5. Enter the following information for the offline depot and click **Save**.

- Hostname
- Port
- Username
- Password

Download VCF Core Component Binaries to an Offline Depot

VMware Cloud Foundation supports an offline depot that you can connect to from multiple VCF Instances. Use the VCF Download Tool to download binaries for the VCF core components to the offline depot.

- Generate a download token on the Broadcom Support Portal.
- [Set Up an Offline Depot Web Server for VMware Cloud Foundation](#).
- Connect to the offline depot. See [Connect to an Offline Depot for a VCF Instance](#).
Note: You can also connect to the offline depot after you download binaries to the offline depot.
- The offline depot must have:
 - The latest version of the VCF Download Tool. You can download it from the Broadcom Support portal.
 - Internet connectivity (either directly or through a proxy) for downloading the binaries and other required files.
 - Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VCF Download Tool for long-running operations.

If the computer hosting the offline depot can only access the Internet using a proxy server, use the following options when downloading:

Option	Description
--proxy-server	Provide the proxy server FQDN and port. For example: --proxy-server proxy.example.com:3128.
--proxy-https	Add this option if the proxy server uses HTTPS. To use this option, the proxy certificate must be added to Bundle Transfer Utility JRE default trust store. For example: /opt/obtu/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore /opt/obtu/jre/lin64/lib/security/cacerts
--proxy-user	For a proxy server that requires authentication, enter the user name.
--proxy-user-password-file	For a proxy server that requires authentication, enter the path to a file where the password for proxy authentication is stored. The file content is used as the proxy password. For example, --proxy-user-password-file ../../password.txt.

Note: VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

You can also use the VCF Download Tool to download binaries for VCF management components and to download ESX components. This procedure focuses on downloading VCF core component binaries.

1. On the computer hosting offline depot, download the binaries to the offline depot root directory.

For example:

```
./vcf-download-tool binaries download --depot-store offline_depot_root_dir --depot-download-token-file token_file_path --vcf-version="9.0.0.0" --lifecycle-managed-by SDDC_MANAGER_VCF --type UPGRADE
```

Variable	Replace With
offline_depot_root_dir	The offline depot root directory.
token_file_path	The path to a text file containing single line with your depot download token. You can generate a download token in the Broadcom Support Portal.

This command downloads the upgrade/patch binaries for VCF 9.0 core components. Specify **INSTALL** as the --type to download install binaries.

2. For information about additional binaries commands and options, see [VCF Download Tool Binaries Commands and Options](#).

Download VCF Core Component Binaries From an Online or Offline Depot

After you connect SDDC Manager to an online or offline depot, you can download upgrade, patch, and install binaries and use those binaries to update or deploy VCF domains.

Connect to an online or offline depot.

- [Connect to an Online Depot for a VCF Instance](#)
- [Connect to an Offline Depot for a VCF Instance](#)

If you are connected to an offline depot, download the binaries to the offline depot. See [Download VCF Core Component Binaries to an Offline Depot](#).

1. In VCF Operations, click **Fleet Management > Lifecycle**.
2. Click **VCF Instances** and click the name of a VCF Instance.
3. Click **Binary Management**.
4. Select the **VCF Version** from the drop-down menu.
5. Click to select the type of binaries you want to download.

- **Upgrade Binaries**
- **Patch Binaries**
- **Install Binaries**

6. Select one or more binaries to download and select an option.

Option	Details
Download	Select this option to start downloading binaries now.
Schedule Download	Select this option to schedule a date and time to start downloading binaries.

Download Upgrade Binaries for All Core Components Without an Online or Offline Depot

If you do not have access to an online or offline depot, you can use the VCF Download Tool to download the binaries to a different computer and then upload them to the SDDC Manager appliance.

- A Windows or Linux computer with Internet connectivity (either directly or through a proxy) for downloading the binaries.
- Configure TCP keepalive in your SSH client to prevent socket connection timeouts for long-running operations.

Note: The VCF Download Tool is the only supported method for downloading binaries. Do not use third-party tools or other methods to download binaries.

Using the VCF Download Tool to download VCF core component binaries involves the following steps:

- Download the latest version of the tool.
- Generate a download token on the Broadcom Support Portal.
- On a computer with access to the Internet, download the binaries.
- Copy the binaries to the SDDC Manager appliance.
- On the SDDC Manager appliance, upload the binaries and other required files to the internal LCM repository.

If the computer with Internet access can only access the Internet using a proxy server, use the proxy server options as described in [VCF Download Tool Binaries Commands and Options](#).

Note: VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

1. Download the most recent version of the VCF Download Tool on a computer with Internet access.
 - a) Log in to the Broadcom Support Portal, click **My Downloads**, and browse to the VMware Cloud Foundation downloads.
 - b) Click your version of VMware Cloud Foundation.
 - c) Click **Drivers & Tools**.
 - d) Click the download icon for the VCF Download Tool.
 - e) Extract `vcf-download-tool-<version.build>.tar.gz`.
 - f) Navigate to the `vcf-download-tool-<version.build>/bin/` and confirm that you have execute permission on all folders.
2. Download the binaries for VCF core components.

```
./vcf-download-tool binaries download --depot-store binaries_dir --depot-download-token-file token_file_path --vcf-version x.x.x.x --lifecycle-managed-by SDDC_MANAGER_VCF
```

<code>binaries_dir</code>	Path to the directory where the binaries should be downloaded. This directory must have read and write permissions. If you do not specify the download directory, binaries are downloaded to the default directory with read and write permissions.
<code>token_file_path</code>	Path to a text file that contains a single line with your download token from the Broadcom Support Portal.
<code>x.x.x.x</code>	The version of VMware Cloud Foundation for which you want to download upgrade/patch binaries. For example: 9.0.0.0.

3. Copy the entire binaries directory to the SDDC Manager appliance.

If you have direct access to the SDDC Manager appliance, use `rsync`. Alternatively, you must copy the files to a portable media and connect the media to the appliance.

Replace `binaries_dir` with the directory to which you downloaded the SDDC Manager appliance upgrade bundle. For example, `/var/opt/obtu`.

Replace `sddc_binaries_dir` with any location on the SDDC Manager appliance that has enough free space available. For example, use `/nfs/vmware/vcf/nfs-mount/upload`.

```
rsync -aP binaries_dir vcf@sddc-manager:sddc_binaries_dir
```

4. Copy the VCF Download Tool to the SDDC Manager appliance.
 - a) SSH in to the SDDC Manager appliance using the `vcf` user account.
 - b) Enter `su` to switch to the root user.
 - c) Create the `lcm-tools` directory.

```
mkdir /home/vcf/lcm-tools
```

Note: If the `/home/vcf/lcm-tools` directory already exists with an older version of the VCF Download Tool, you need to delete contents of the existing directory before proceeding.

- d) Copy the VCF Download Tool file (`vcf-download-tool-<version.build>.tar.gz`) that you downloaded to the `/home/vcf/lcm-tools` directory.
- e) Extract the contents of `vcf-download-tool-<version.build>.tar.gz`.

```
tar -xvf vcf-download-tool-<version.build>.tar.gz
```

5. From the SDDC Manager appliance, use the VCF Download Tool to upload the binaries.

```
./vcf-download-tool binaries upload --depot-store binaries_dir --sddc-manager-fqdn FQDN --sddc-manager-user username --sddc-manager-user-password-file password_file_path
```

Variable	Replace With
<code>binaries_dir</code>	The directory on the SDDC Manager appliance where you copied all the binaries and metadata downloaded by the VCF Download Tool.
<code>FQDN</code>	The Fully Qualified Domain Name (FQDN) of the SDDC Manager. This is the address that the VCF Download Tool will use to connect to SDDC Manager.
<code>username</code>	The user name for connecting to the SDDC Manager.
<code>password_file_path</code>	The path to a text file containing the password for the SDDC Manager user name. The password should be on a single line within the file. If this file is not provided, you are prompted to manually enter the password during execution.

Download Upgrade Binaries for a Subset of Core Components Without an Online or Offline Depot

VCF Operations includes an upgrade planner for VCF domains that allows you to select specific target versions for each VCF core component you want to upgrade or update. You use the VCF Download Tool to download the binaries when SDDC Manager is not connected to an online or offline depot.

- Generate a download token on the Broadcom Support Portal.
- A Windows or Linux computer with Internet connectivity (either directly or through a proxy) for downloading the binaries.
- The latest version of the VCF Download Tool available on both the computer with Internet access and the SDDC Manager appliance. You can download the VCF Download Tool from the Broadcom Support Portal.
- Configure TCP keepalive in your SSH client to prevent socket connection timeouts for long-running operations.

Note: The VCF Download Tool is the only supported method for downloading binaries. Do not use third-party tools or other methods to download binaries.

You can use the upgrade planner in VCF Operations to select any supported version for each of the VCF core components. This includes patch versions as well as VCF BOM versions.

Downloading flexible BOM upgrade bundles involves the following steps:

- On a computer with access to the Internet, use the VCF Download Tool to download the metadata files for the VCF core component binaries.

- Copy the metadata files to the SDDC Manager appliance.
- On the SDDC Manager appliance, use the VCF Download Tool to upload the metadata files.
- Plan a VCF domain upgrade using VCF Operations.
- On the SDDC Manager appliance, use the VCF Download Tool to generate a download specification file.
- Copy the download specification file to the computer with Internet access.
- On the computer with access to the Internet, download binaries using the download specification file.
- Copy the binary directory to the SDDC Manager appliance and use the VCF Download Tool to upload the binaries.

If the computer with Internet access can only access the Internet using a proxy server, use the proxy server options as described in [VCF Download Tool Metadata Commands and Options](#) and [VCF Download Tool Binaries Commands and Options](#).

Note: VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

1. On the computer with Internet access, download the metadata files.

```
./vcf-download-tool metadata download --depot-store metadata_dir --depot-download-token-file token_file_path
```

Variable	Replace With
<i>metadata_dir</i>	The directory path where all metadata downloaded by the VCF Download Tool will be stored.
<i>token_file_path</i>	The path to a text file containing single line with your depot download token. You can generate a download token in the Broadcom Support Portal.

2. Copy the entire directory containing the downloaded metadata to the SDDC Manager appliance.

3. On the SDDC Manager appliance, upload the metadata.

```
./vcf-download-tool metadata upload --depot-store metadata_dir--sddc-manager-fqdn FQDN --sddc-manager-user username --sddc-manager-user-password-file password_file_path
```

Variable	Replace With
<i>metadata_dir</i>	The directory on the SDDC Manager appliance where you copied all the metadata downloaded by the VCF Download Tool.
<i>FQDN</i>	The Fully Qualified Domain Name (FQDN) of the SDDC Manager. This is the address that the VCF Download Tool will use to connect to SDDC Manager.
<i>username</i>	The user name for connecting to the SDDC Manager.
<i><password_file_path</i>	The path to a text file containing the password for the SDDC Manager user name. The password should be on a single line within the file. If this file is not provided, you are prompted to manually enter the password during execution.

4. Plan the upgrade in VCF Operations.

See [Upgrade or Update the Core Components in your VCF Domain](#).

5. On the SDDC Manager appliance, use the VCF Download Tool to generate a download specification file.

```
./vcf-download-tool binaries create-download-spec --domain-name domain_name --download-spec-dir spec_output_dir --sddc-manager-fqdn FQDN --sddc-manager-user username --sddc-manager-user-password-file password_file_path
```

Variable	Replace With
<i>domain_name</i>	One or more VCF domain names. Use comma separation for multiple domains.
<i>spec_output_dir</i>	The directory on the SDDC Manager appliance in which to save the download specification file.

Variable	Replace With
<i>FQDN</i>	The Fully Qualified Domain Name (FQDN) of the SDDC Manager. This is the address that the VCF Download Tool will use to connect to SDDC Manager.
<i>username</i>	The user name for connecting to the SDDC Manager.
<i>password_file_path</i>	The path to a text file containing the password for the SDDC Manager user name. The password should be on a single line within the file. If this file is not provided, you are prompted to manually enter the password during execution.

6. Copy download specification file to the computer with Internet access.

7. On the computer with access to the Internet, download binaries using the download specification file.

```
./vcf-download-tool binaries download --depot-store binaries_dir --depot-download-token-file token_file_path --download-spec-file download_spec_file_path
```

<i>binaries_dir</i>	Path to the directory where the binaries should be downloaded. This directory must have read and write permissions. If you do not specify the download directory, binaries are downloaded to the default directory with read and write permissions.
<i>token_file_path</i>	Path to a text file that contains a single line with your download token from the Broadcom Support Portal.
<i>download_spec_file_path</i>	Path to the download specification file.

8. Copy the entire binaries directory to the SDDC Manager appliance.

If you have direct access to the SDDC Manager appliance, use `rsync`. Alternatively, you must copy the files to a portable media and connect the media to the appliance.

Replace *binaries_dir* with the directory to which you downloaded the SDDC Manager appliance upgrade bundle. For example, `/var/opt/obtu`.

Replace *sddc_binaries_dir* with any location on the SDDC Manager appliance that has enough free space available. For example, use `/nfs/vmware/vcf/nfs-mount/upload`.

```
rsync -aP binaries_dir vcf@sddc-manager:sddc_binaries_dir
```

9. On the SDDC Manager appliance, upload the binaries.

```
./vcf-download-tool binaries upload --depot-store binaries_dir --sddc-manager-fqdn FQDN --sddc-manager-user username --sddc-manager-user-password-file password_file_path
```

Variable	Replace With
<i>binaries_dir</i>	The directory on the SDDC Manager appliance where you copied all the binaries.
<i>FQDN</i>	The Fully Qualified Domain Name (FQDN) of the SDDC Manager. This is the address that the VCF Download Tool will use to connect to SDDC Manager.
<i>username</i>	The user name for connecting to the SDDC Manager.
<i>password_file_path</i>	The path to a text file containing the password for the SDDC Manager user name. The password should be on a single line within the file. If this file is not provided, you are prompted to manually enter the password during execution.

Using the VCF Download Tool

The VCF Download Tool is a command-line interface (CLI) utility that is designed to simplify the management of binaries and metadata for VMware Cloud Foundation platforms.

The VCF Download Tool provides a suite of commands for downloading, uploading, listing, and cleaning up binaries, enabling efficient management of software components for VMware Cloud Foundation releases. The tool also supports the installation, uninstallation, and execution of the Update Manager Download Service (UMDS). With its comprehensive features, the VCF Download Tool helps streamline the process of preparing, managing, and maintaining VCF-related artifacts in your environment.

VCF Download Tool General Options

You can use the VCF Download Tool general options to view help, check the utility version, and choose whether or not to share data with the Customer Experience Improvement Program (CEIP).

Help Options

Option	Description
<code>-h, --help</code>	Provides information about the VCF Download Tool options and commands.
<code>binaries -h, binaries --help</code>	Provides information about the options for managing binaries.
<code>metadata -h, metadata --help</code>	Provides information about the options for managing metadata.
<code>releases -h, releases --help</code>	Provides information about the options for VMware Cloud Foundation releases.
<code>umds -h, umds --help</code>	Provides information about the options for the Update Manager Download Service (UMDS).

Example:

```
./vcf-download-tool binaries -h
```

Version Option

Option	Description
<code>-v, --version</code>	Displays the version of the VCF Download Tool.

Customer Experience Improvement Program (CEIP) Option

The VCF Download Tool participates in the VMware Customer Experience Improvement Program (CEIP). Under CEIP, VMware collects information about participating customers' use of on-premise products. You can enable or deactivate CEIP for the VCF Download Tool.

For additional information regarding the CEIP, see <https://www.broadcom.com/company/legal/privacy/data-usage-programs/ceip>.

Option	Description
<code>--ceip ENABLE</code> or <code>--ceip DISABLE</code>	The first time you use the VCF Download Tool, you are prompted to make a choice about whether to participate in CEIP. This selection is maintained and used as the default, unless you specify the option again with a different value.

VCF Download Tool Binaries Commands and Options

Use the VCF Download Tool binaries commands to list, download, upload, and manage VMware Cloud Foundation installation, upgrade, and patch binaries.

Binaries Commands

Command	Description
<code>download</code>	Downloads binaries and metadata (unified release manifest, product version catalog, vSAN HCL, and compatibility data). The command supports filtering to narrow down the selection. At least one filter is required. When executed multiple times with the same output directory, the command's behavior is cumulative, adding to previously downloaded binaries without overwriting them. Note: The <code>download</code> command does not include UMDS binaries.
<code>upload</code>	Uploads downloaded binaries and metadata into SDDC Manager. Before uploading, you must transfer the downloaded binaries and metadata to the SDDC Manager appliance. The <code>upload</code> command must be run from within the SDDC Manager appliance.
<code>list</code>	Provides a comprehensive view of all available binaries. The command supports filtering to narrow down the selection. At least one filter is required. The <code>list</code> command displays detailed information for each binary, including the component name, version number, release date, size, and type. This makes it easier to identify which binaries are relevant to your requirements before downloading.
<code>cleanup</code>	Use the <code>cleanup</code> command to manage disk space by removing unwanted or obsolete binaries, while retaining the ones you still need. Run the <code>cleanup</code> command on a system where you have already downloaded binaries.
<code>create-download-spec</code>	Generates a download specification file based on the planned upgrades for a specific domain within the environment. The specification file is used to identify which binaries are required for the planned upgrades. After you create the specification file, you can use it as an input parameter for the <code>download</code> command, ensuring that only the relevant binaries for the planned upgrade are downloaded. The <code>create-download-spec</code> command must be run from within the SDDC Manager appliance.

Binaries Download Options

If the computer with Internet access can only access the Internet using a proxy server, use the following options when downloading binaries:

Table 916: Proxy Server Options

Option	Description
<code>--proxy-https</code>	Add this option if the proxy server uses HTTPs.
<code>--proxy-user-password-file proxyServerPasswordFile</code>	For a proxy server that requires authentication, replace <i>proxyServerPasswordFile</i> with the path to a file where the password for proxy authentication is stored. The file content is used as the proxy password. For example, <code>--proxyPasswordFile ../../password.txt</code> . Note: The supported proxy authentication methods are Basic and NTLMv1.
<code>-r proxyServerUserName</code> or <code>--proxy-user proxyServerUserName</code>	For a proxy server that requires authentication, replace <i>proxyServerUserName</i> with the user name.
<code>-s proxyServerHostName</code> or <code>--proxy-server proxyServerHostName</code>	Replace <i>proxyServerHostName</i> with the proxy server FQDN and port. For example: <code>--proxy-server proxy.example.com:3128</code> .

Table 917: Depot Options

Option	Description
<code>-d depotStore</code> or <code>--depot-store depotStore</code>	Directory where all binaries and metadata downloaded by the VCF Download Tool will be stored. Replace <i>depotStore</i> with the directory path. This directory serves as the local repository for all downloaded artifacts, maintaining the necessary structure for easy access and management.
<code>--depot-download-token-file depotDownloadTokenFile</code>	Replace <i>depotDownloadTokenFile</i> with the path to a text file containing single line with your depot download token. You can generate a download token in the Broadcom Support Portal.

Table 918: Filter Options

Option	Description
<code>--automated-install</code>	Downloads the binaries needed by the VCF Installer.
<code>--component component</code>	Downloads the binaries for a specified component. Replace <i>component</i> with one of the following values: - VCENTER - SDDC_MANAGER_VCF - NSX_T_MANAGER - ESX_HOST - VRSLCM - VRA - VROPS - VRLI

Option	Description
	- VRNI - VSAN_OSA_WITNESS - VSAN_ESA_WITNESS - VSAN_FILE_SERVICES - VMTOOLS - VCFDT - VCF_OPS_CLOUD_PROXY - VIDB - HCX - VLR - DLVM - VMRC - VRO
<code>--lifecycle-managed-by lcm</code>	Downloads binaries whose lifecycle is managed by the specified component. Replace <i>lcm</i> with one of the following values: - SDDC_MANAGER_VCF - VRSLCM - SELF
<code>--sku sku</code>	Downloads binaries for VMware Cloud Foundation or VMware vSphere Foundation. Replace <i>sku</i> with one of the following values: - VCF - VVF
<code>-t type</code> or <code>--type type</code>	Downloads binaries of the specified type. Replace <i>type</i> with one of the following values: - INSTALL - UPGRADE
<code>--vcf-version version[..<i>end version</i>]</code>	Downloads binaries for a specific version <i>version</i> of VMware Cloud Foundation, or for a range of versions <i>start_version</i> .. <i>end_version</i> . You can enter the version string in the following formats: <i>a.b</i>: For example, if you enter 9.0, the tool downloads the binaries for 9.0.0.0 and any 9.0.x binaries. <i>a.b.c</i>: For example, if you enter 9.0.1, the tool downloads all 9.0.1.x binaries. <i>a.b.c.d</i>: For example, if you enter 9.0.1.0, the tool only downloads 9.0.1.0 binaries. When you specify a version range, the tool downloads all binaries from the start version to the end version and includes the binaries for both the start and end versions.
<code>--id bundleIds</code>	Downloads specific binaries using the bundle ID(s). Replace <i>bundleIds</i> with a single bundle ID or a comma-separated list of multiple bundle IDs.
<code>--download-spec-file downloadSpecFile</code>	Downloads only the binaries required for a specific planned upgrade. Replace <i>downloadSpecFile</i> with the path to the download specification file generated by the <code>create-download-spec</code> command. This file contains the details of the binaries required for the planned upgrade.

Table 919: Binaries Download Examples

Example Command	Description
<code>./vcf-download-tool binaries download --depot-store <i>binaries_dir</i> --depot-download-token-file <i>token_file_path</i> --vcf-version=9.0.0.0 --sku VCF --automated-install --type INSTALL</code>	Downloads the install binaries needed by VCF Installer for deploying a VMware Cloud Foundation 9.0.
<code>./vcf-download-tool binaries download --depot-store <i>binaries_dir</i> --depot-download-token-file <i>token_file_path</i> --download-spec-file <i>download_spec_file_path</i> --proxy-server <i>FQDN:Port</i> --proxy-https --proxy-user <i>username</i> --proxy-user-password-file <i>password_file_path</i></code>	Download binaries based on a download specification file using an authenticated proxy server.

Binaries Upload Options

Option	Description
<code>--depot-store <i>depotStore</i> or -d <i>depotStore</i></code>	The directory on the SDDC Manager appliance where you copied all the binaries and metadata downloaded by the VCF Download Tool.
<code>-f <i>sddcManagerFqdn</i> or --sddc-manager-fqdn <i>sddcManagerFqdn</i></code>	Replace <i>sddcManagerFqdn</i> with the Fully Qualified Domain Name (FQDN) of the SDDC Manager. This is the address that the VCF Download Tool will use to connect to the SDDC Manager instance.
<code>-u <i>sddcManagerUsername</i> or --sddc-manager-user <i>sddcManagerUsername</i></code>	Replace <i>sddcManagerUsername</i> with the user name for connecting to the SDDC Manager.
<code>--sddc-manager-user-password-file <i>sddcManagerPasswordFile</i></code>	Replace <i>sddcManagerPasswordFile</i> with the path to a text file containing the password for the provided <code>--sddc-manager-user</code> . The password should be on a single line within the file. If this file is not provided, you are prompted to manually enter the password during execution.

For example:

```
./vcf-download-tool binaries upload --depot-store= binaries_dir --sddc-manager-fqdn FQDN --sddc-manager-user username --sddc-manager-user-password-file password_file_path
```

Binaries List Options

If the computer with Internet access can only access the Internet using a proxy server, use the following options when listing binaries:

Table 920: Proxy Server Options

Option	Description
<code>--proxy-https</code>	Add this option if the proxy server uses HTTPs.

Option	Description
<code>--proxy-user-password-file <i>proxyServerPasswordFile</i></code>	For a proxy server that requires authentication, replace <i>proxyServerPasswordFile</i> with the path to a file where the password for proxy authentication is stored. The file content is used as the proxy password. For example, <code>--proxyPasswordFile ../../password.txt</code> . Note: The supported proxy authentication methods are Basic and NTLMv1.
<code>-r <i>proxyServerUserName</i> or --proxy-user <i>proxyServerUserName</i></code>	For a proxy server that requires authentication, replace <i>proxyServerUserName</i> with the user name.
<code>-s <i>proxyServerHostName</i> or --proxy-server <i>proxyServerHostName</i></code>	Replace <i>proxyServerHostName</i> with the proxy server FQDN and port. For example: <code>--proxy-server proxy.example.com:3128</code> .

Table 921: Depot Options

Option	Description
<code>--depot-download-token-file <i>depotDownloadTokenFile</i></code>	Replace <i>depotDownloadTokenFile</i> with the path to a text file containing single line with your depot download token. You can generate a download token in the Broadcom Support Portal.

Table 922: Filter Options

<code>--automated-install</code>	Lists the binaries needed by the VCF Installer.
<code>--component <i>component</i></code>	Lists the binaries for a specified component. Replace <i>component</i> with one of the following values: - VCENTER - SDDC_MANAGER_VCF - NSX_T_MANAGER - ESX_HOST - VRSLCM - VRA - VROPS - VRLI - VRNI - VSAN_OSA_WITNESS - VSAN_ESA_WITNESS - VSAN_FILE_SERVICES - VMTOOLS - VCFDT - VCF_OPS_CLOUD_PROXY - VIDB - HXC

	- VLR - DLVM - VMRC - VRO
<code>--lifecycle-managed-by <i>lcm</i></code>	Lists binaries whose lifecycle is managed by the specified component. Replace <i>lcm</i> with one of the following values: - SDDC_MANAGER_VCF - VRSLCM - SELF
<code>--patches-only</code>	Lists the binaries for patch versions of components.
<code>--sku <i>sku</i></code>	Lists binaries for VMware Cloud Foundation or VMware vSphere Foundation. Replace <i>sku</i> with one of the following values: - VCF - VVF
<code>-t <i>type</i></code> or <code>--type <i>type</i></code>	Lists binaries of the specified type. Replace <i>type</i> with one of the following values: - INSTALL - UPGRADE
<code>--vcf-version <i>version</i>[..<i>end version</i>]</code>	Lists binaries for a specific version <i>version</i> of VMware Cloud Foundation, or for a range of versions <i>start_version</i> .. <i>end_version</i> . You can enter the version string in the following formats: <i>a.b</i>: For example, if you enter 9.0, the tool lists the binaries for 9.0.0.0 and any 9.0.x binaries. <i>a.b.c</i>: For example, if you enter 9.0.1, the tool lists all 9.0.1.x binaries. <i>a.b.c.d</i>: For example, if you enter 9.0.1.0, the tool only lists 9.0.1.0 binaries. When you specify a version range, the tool lists all binaries from the start version to the end version and includes the binaries for both the start and end versions.
<code>--id <i>bundleIds</i></code>	Lists specific binaries using the bundle ID(s). Replace <i>bundleIds</i> with a single bundle ID or a comma-separated list of multiple bundle IDs.
<code>--download-spec-file <i>downloadSpecFile</i></code>	Lists only the binaries required for a specific planned upgrade. Replace <i>downloadSpecFile</i> with the path to the download specification file generated by the <code>create-download-spec</code> command. This file contains the details of the binaries required for the planned upgrade.

Example:

```
./vcf-download-tool binaries list --depot-download-token-file token_file_path --vcf-version 9.0.0.0 --sku VCF --automated-install --type INSTALL
```

Binaries Cleanup Options

Table 923: Depot Options

Option	Description
<code>-d <i>depotStore</i></code> or <code>--depot-store <i>depotStore</i></code>	Directory where all binaries and metadata downloaded by the VCF Download Tool are stored. Replace <i>depotStore</i> with the directory path. This directory serves as the local repository for all downloaded artifacts, maintaining the necessary structure for easy access and management.

Table 924: Filter Options

Option	Description
<code>--automated-install</code>	Cleans up the binaries needed by the VCF Installer.
<code>--component <i>component</i></code>	Cleans up the binaries for a specified component. Replace <i>component</i> with one of the following values: - VCENTER - SDDC_MANAGER_VCF - NSX_T_MANAGER - ESX_HOST - VRSLCM - VRA - VROPS - VRLI - VRNI - VSAN_OSA_WITNESS - VSAN_ESA_WITNESS - VSAN_FILE_SERVICES - VMTOOLS - VCFDT - VCF_OPS_CLOUD_PROXY - VIDB - HCX - VLR - DLVM - VMRC - VRO
<code>--lifecycle-managed-by <i>lcm</i></code>	Cleans up binaries whose lifecycle is managed by the specified component. Replace <i>lcm</i> with one of the following values: - SDDC_MANAGER_VCF - VRSLCM - SELF
<code>--patches-only</code>	Cleans up the binaries for patch versions of components.
<code>--sku <i>sku</i></code>	Cleans up binaries for VMware Cloud Foundation or VMware vSphere Foundation. Replace <i>sku</i> with one of the following values: VCF

Option	Description
<code>-t type</code> or <code>--type type</code>	- VVF Cleans up binaries of the specified type. Replace <i>type</i> with one of the following values: - INSTALL - UPGRADE
<code>--vcf-version version[..<i>end version</i>]</code>	Cleans up binaries for a specific version <i>version</i> of VMware Cloud Foundation, or for a range of versions <i>start_version</i>..<i>end_version</i>. You can enter the version string in the following formats: <i>a.b</i>: For example, if you enter 9.0, the tool cleans up the binaries for 9.0.0.0 and any 9.0.x binaries. <i>a.b.c</i>: For example, if you enter 9.0.1, the tool cleans up all 9.0.1.x binaries. <i>a.b.c.d</i>: For example, if you enter 9.0.1.0, the tool only cleans up 9.0.1.0 binaries. When you specify a version range, the tool cleans up all binaries from the start version to the end version and includes the binaries for both the start and end versions.
<code>--id bundleIds</code>	Cleans up specific binaries using the bundle ID(s). Replace <i>bundleIds</i> with a single bundle ID or a comma-separated list of multiple bundle IDs.
<code>--download-spec-file downloadSpecFile</code>	Cleans up only the binaries required for a specific planned upgrade. Replace <i>downloadSpecFile</i> with the path to the download specification file generated by the <code>create-download-spec</code> command. This file contains the details of the binaries required for the planned upgrade.

For example:

```
./vcf-download-tool binaries cleanup --depot-store binaries_dir
--vcf-version VCF_version
--lifecycle-managed-by SDDC_MANAGER_VCF --type UPGRADE
```

Binaries Create Download Spec Options

Option	Description
<code>--domain-name domainNames</code>	Replace <i>domainNames</i> with a single workload domain or multiple comma-separated workload domain names.
<code>--download-spec-dir downloadSpecFile</code>	Replace <i>downloadSpecFile</i> with the path to the directory where the file generated by the <code>create download spec</code> command will be stored.
<code>-f sddcManagerFqdn</code> or <code>--sddc-manager-fqdn sddcManagerFqdn</code>	Replace <i>sddcManagerFqdn</i> with the Fully Qualified Domain Name (FQDN) of the SDDC Manager. This is the address that the VCF Download Tool will use to connect to the SDDC Manager instance.
<code>-u sddcManagerUsername</code> or <code>--sddc-manager-user sddcManagerUsername</code>	Replace <i>sddcManagerUsername</i> with the user name for connecting to the SDDC Manager.

Option	Description
<code>--sddc-manager-user-password-file sddcManagerPasswordFile</code>	Replace <i>sddcManagerPasswordFile</i> with the path to a text file containing the password for the provided <code>--sddc-manager-user</code> . The password should be on a single line within the file. If this file is not provided, you are prompted to manually enter the password during execution.

Example:

```
./vcf-download-tool binaries create-download-spec --domain-name domain_name
--download-spec-dir spec_output_dir
--sddc-manager-fqdn FQDN
--sddc-manager-user username
--sddc-manager-user-password-file password_file_path
```

VCF Download Tool Metadata Commands and Options

Use the VCF Download Tool metadata commands to download and upload the metadata files (unified release manifest, product version catalog, vSAN HCL, and compatibility data) for VMware Cloud Foundation binaries.

Metadata Commands

Command	Description
download	Downloads all the metadata required by SDDC Manager in a single operation. It automatically retrieves all relevant metadata files, without requiring you to manually specify which files are needed. Note: The <code>download</code> command does not include UMDS metadata.
upload	Uploads all the downloaded metadata to SDDC Manager. Before uploading, you must transfer the downloaded metadata to the SDDC Manager appliance. The <code>upload</code> command must be run from within the SDDC Manager appliance.

Metadata Download Options

If the computer with Internet access can only access the Internet using a proxy server, use the following options when downloading metadata:

Table 925: Proxy Server Options

Option	Description
<code>--proxy-https</code>	Add this option if the proxy server uses HTTPSs.
<code>--proxy-user-password-file proxyServerPasswordFile</code>	For a proxy server that requires authentication, replace <i>proxyServerPasswordFile</i> with the path to a file where the password for proxy authentication is stored. The file content is used as the proxy password.

Option	Description
	For example, <code>--proxyPasswordFile ../../password.txt</code> . Note: The supported proxy authentication methods are Basic and NTLMv1.
<code>-r proxyServerUserName</code> or <code>--proxy-user proxyServerUserName</code>	For a proxy server that requires authentication, replace <i>proxyServerUserName</i> with the user name.
<code>-s proxyServerHostName</code> or <code>--proxy-server proxyServerHostName</code>	Replace <i>proxyServerHostName</i> with the proxy server FQDN and port. For example: <code>--proxy-server proxy.example.com:3128</code> .

Option	Description
<code>-d depotStore</code> or <code>--depot-store depotStore</code>	Directory where all binaries and metadata downloaded by the VCF Download Tool will be stored. Replace <i>depotStore</i> with the directory path. This directory serves as the local repository for all downloaded artifacts, maintaining the necessary structure for easy access and management.
<code>--depot-download-token-file depotDownloadTokenFile</code>	Replace <i>depotDownloadTokenFile</i> with the path to a text file containing single line with your depot download token. You can generate a download token in the Broadcom Support Portal.

Example:

```
./vcf-download-tool metadata download --depot-store metadata_dir --depot-download-token-file token_file_path
```

Metadata Upload Options

Option	Description
<code>--depot-store depotStore</code> or <code>-d depotStore</code>	The directory on the SDDC Manager appliance where you copied all the metadata downloaded by the VCF Download Tool.
<code>-f sddcManagerFqdn</code> or <code>--sddc-manager-fqdn sddcManagerFqdn</code>	Replace <i>sddcManagerFqdn</i> with the Fully Qualified Domain Name (FQDN) of the SDDC Manager. This is the address that the VCF Download Tool will use to connect to the SDDC Manager instance.
<code>-u sddcManagerUsername</code> or <code>--sddc-manager-user sddcManagerUsername</code>	Replace <i>sddcManagerUsername</i> with the user name for connecting to the SDDC Manager.
<code>--sddc-manager-user-password-file sddcManagerPasswordFile</code>	Replace <i>sddcManagerPasswordFile</i> with the path to a text file containing the password for the provided <code>--sddc-manager-user</code> . The password should be on a single line within the file. If this file is not provided, you are prompted to manually enter the password during execution.

Example:

```
./vcf-download-tool metadata upload --depot-store metadata_dir
--sddc-manager-fqdn FQDN>
```

```
--sddc-manager-user username
--sddc-manager-user-password-file password_file_path
```

VCF Download Tool Releases Commands

Use the VCF Download Tool releases commands to download the unified release manifest and view a list of available VMware Cloud Foundation releases.

Releases Commands

Command	Description
<code>list</code>	Provides a comprehensive list of all available VMware Cloud Foundation releases. The <code>list</code> command supports filtering to narrow down the selection. When executed without specifying a release, the <code>list</code> command displays all the releases currently available. If a specific release is provided as a parameter, the command shows detailed information about the products included within that release.

Releases Options

If the computer with Internet access can only access the Internet using a proxy server, use the following options when listing releases:

Table 926: Proxy Server Options

Option	Description
<code>--proxy-https</code>	Add this option if the proxy server uses HTTPS.
<code>--proxy-user-password-file proxyServerPasswordFile</code>	For a proxy server that requires authentication, replace <i>proxyServerPasswordFile</i> with the path to a file where the password for proxy authentication is stored. The file content is used as the proxy password. For example, <code>--proxyPasswordFile ../../password.txt</code> . Note: The supported proxy authentication methods are Basic and NTLMv1.
<code>-r proxyServerUserName</code> or <code>--proxy-user proxyServerUserName</code>	For a proxy server that requires authentication, replace <i>proxyServerUserName</i> with the user name.
<code>-s proxyServerHostName</code> or <code>--proxy-server proxyServerHostName</code>	Replace <i>proxyServerHostName</i> with the proxy server FQDN and port. For example: <code>--proxy-server proxy.example.com:3128</code> .

Table 927: Depot Options

Option	Description
<code>--depot-download-token-file <i>depotDownloadTokenFile</i></code>	Replace <i>depotDownloadTokenFile</i> with the path to a text file containing single line with your depot download token. You can generate a download token in the Broadcom Support Portal.

Table 928: Filter Options

Option	Description
<code>--vcf-version <i>version</i></code>	Replace <i>version</i> with the VMware Cloud Foundation version for which to show detailed information about the products included within that release. If you do not provide a version, the <code>list</code> command returns all available versions.

Example:

```
./vcf-download-tool releases list --vcf-version 9.0.0.0 --depot-download-token-file token_file_path
```

VCF Download Tool Update Manager Download Service (UMDS) Commands

Use the VCF Download Tool UMDS commands to install UMDS and list, download and manage ESX binaries and metadata. You can use this data to create vSphere Lifecycle Manager images in the vSphere Client.

UMDS Commands

Command	Description
<code>install</code>	Installs the UMDS (Update Manager Download Service) tool, which is required to run UMDS-specific operations. Once installed, the UMDS tool enables you to execute various tasks related to UMDS. Example: <code>./vcf-download-tool umds install</code>
<code>uninstall</code>	Removes the UMDS tool from the system. Running this command uninstalls UMDS, disabling any further UMDS-specific operations. This command is useful for cleaning up the environment when UMDS is no longer needed or if it needs to be replaced or updated. Example: <code>./vcf-download-tool umds uninstall</code>
<code>run</code>	After you install UMDS, you can use the <code>run</code> command to configure and run UMDS.

UMDS Run Command Options

Option	Description
<code>-D, --download</code>	Downloads updates based on the current configuration. Can use the optional argument <code>-m, --only-metadata</code> to download metadata only.
<code>-E, --export</code>	Exports all updates that have been downloaded. Can use the optional argument <code>-x, --export-store</code> to specify a destination directory for the export operation. This overrides the configured destination directory.
<code>-I, --import</code>	Imports all the updates from a given source location and merge to the UMDS patch store. Arguments: <code>-s, --import-store</code>: The source patch store for import operation. <code>-f, --offline-bundle</code>: The absolute path to an offline zip bundle file to import.
<code>-R, --re-download</code>	Re-downloads existing updates that may be corrupted and downloads new updates. Use this command only if you suspect that the UMDS patch store is corrupted. Can use the optional argument <code>-m, --only-metadata</code> to download metadata only.
<code>-S, --set-config</code>	Configures UMDS. Arguments: <code>-u URL, --add-url URL</code>: Adds a URL to the configuration for downloading updates. Requires <code>--url-type</code>. <code>-r URL_type, --url-type URL_type</code>: Specify HOST for ESX 6.x. Use with <code>--add-url</code>. <code>-l URL, --remove-url URL</code>: Removes a URL from the configuration. <code>-P patch_store, --patch-store patch_store</code>: Configures a location for storing updates after download. <code>-o export_store, --default-export-store export_store</code>: Configures a location for exporting updates. <code>-p FQDN:port, --proxy FQDN:port</code>: Configures proxy server settings. Use <code>--proxy ""</code> to stop using the proxy server. <code>-Y, --enable-host</code>: Enables ESX host update downloads for all platforms <code>-N, --disable-host</code>: Disables ESX host update downloads for all platforms <code>-e host_platform, --enable-host-platform host_platform</code>: Enables ESX host update downloads for specified platforms. Specify multiple platforms separated by whitespace Note: Use <code>-L, --list-host-platforms</code> to list available host platforms. <code>-d host_platform, --disable-host-platform host_platform</code>: Deactivates ESX host update downloads for specified platforms. Specify multiple platforms separated by whitespace

Option	Description
	<code>-s token, --add-entitlement-token token</code>: Adds default URLs with entitlement token ID.
<code>-G, --get-config</code>	Prints the current UMDS configuration.
<code>-v, --version</code>	Prints the UMDS version.
<code>-i level, --info-level level</code>	Specifies the level of information shown on the console. You can use this option with: <code>-D, --download</code> <code>-E, --export</code> <code>-E, --export</code> Arguments: <code>verbose</code> <code>info</code>
<code>-L, --list-host-platforms</code>	Lists all supported ESX platforms for download.

Example UMDS Commands and Options

Command	Description
<code>./vcf-download-tool umds run -S --add-url https://hostname/index.xml --url-type HOST</code>	Adds a new ESX host patch depot URL.
<code>./vcf-download-tool umds run -S --remove-url https://hostname/index.xml</code>	Removes an ESX host patch depot URL.
<code>./vcf-download-tool umds run --list-host-platforms</code>	Lists all supported platforms for downloading ESX host updates.
<code>./vcf-download-tool umds run -S --enable-host</code>	Enables downloading of ESX host updates.
<code>./vcf-download-tool umds run -S -e embeddedEsx-8.0.0</code>	Enables downloading of only ESXi 8.0.0 host updates.
<code>./vcf-download-tool umds run -S -d embeddedEsx-8.0.0</code>	Deactivates downloading of only ESXi 8.0.0 host updates.
<code>./vcf-download-tool umds run vmware-umds -D</code>	Downloads updates based on the current configuration.
<code>./vcf-download-tool umds run -S --default-export-store F:\UMDS-store</code> <code>./vcf-download-tool umds run -E</code>	Specifies and new export location and exports all updates that have been downloaded to that location.

Lifecycle Management of VCF Management Components

Use the VMware Cloud Foundation Operations fleet management appliance to manage, upgrade, patch, and install the management components in VCF 9.0.

- VCF Operations
- VMware Cloud Foundation Operations fleet management
- VCF Operations for networks
- VCF Operations for logs
- VCF Automation
- VCF Identity Broker

Update Order for Management Components for Maintenance Releases

When you update your VCF environment to a maintenance release version, for example, from 9.0.0.0 to 9.0.1.0 or 9.0.2.0, you first update your management components that operate on the fleet level before you update the core components in your instances. For the management components, follow the specified update order:

1. VCF Operations fleet management appliance
2. VCF Operations
3. Update the remaining components VCF Operations for logs, VCF Automation, VCF Identity Broker, and VCF Operations for networks in your preferred order.

For the core components, start with SDDC Manager, followed by NSX, vCenter, ESX hosts, and vSAN. See [Lifecycle Management of VCF Core Components](#).

Before you can upgrade, patch, or deploy the VCF management components, you must download the binaries to the VCF Operations fleet management appliance. The method that you use to download the binaries depends on how you access the Internet in your environment.

- Online depot: The VCF Operations fleet management appliance connects directly to the online depot and requires a download token.
- Offline depot that is a self-managed web server: The VCF Operations fleet management appliance connect to the Internet through a web server and requires a download token.
- Offline depot that is a local server: The VCF Operations fleet management appliance cannot connect to the Internet but can connect to a computer with Internet access. This depot is ideal for dark site environments.

After downloading binaries, you can proceed with your deployment, upgrade, or patch.

- For deployment, see: [Completing the Deployment of Your Platform](#).
- For upgrading, see: [Import an Existing VMware Aria Automation Instance Into VCF Operations and Upgrade to VCF Automation 9.0](#) or [Import an Existing VMware Aria Operations for Networks Instance Into VCF Operations and Upgrade to VCF Operations for Networks](#), or [Upgrade an Individual VCF Management Component](#).
- For patching, see [Patch an Individual VCF Management Component](#).

Downloading VCF Management Components Binaries

Binary management provides a centralized interface that administrators use to download component binaries for upgrades, patches, and installations.

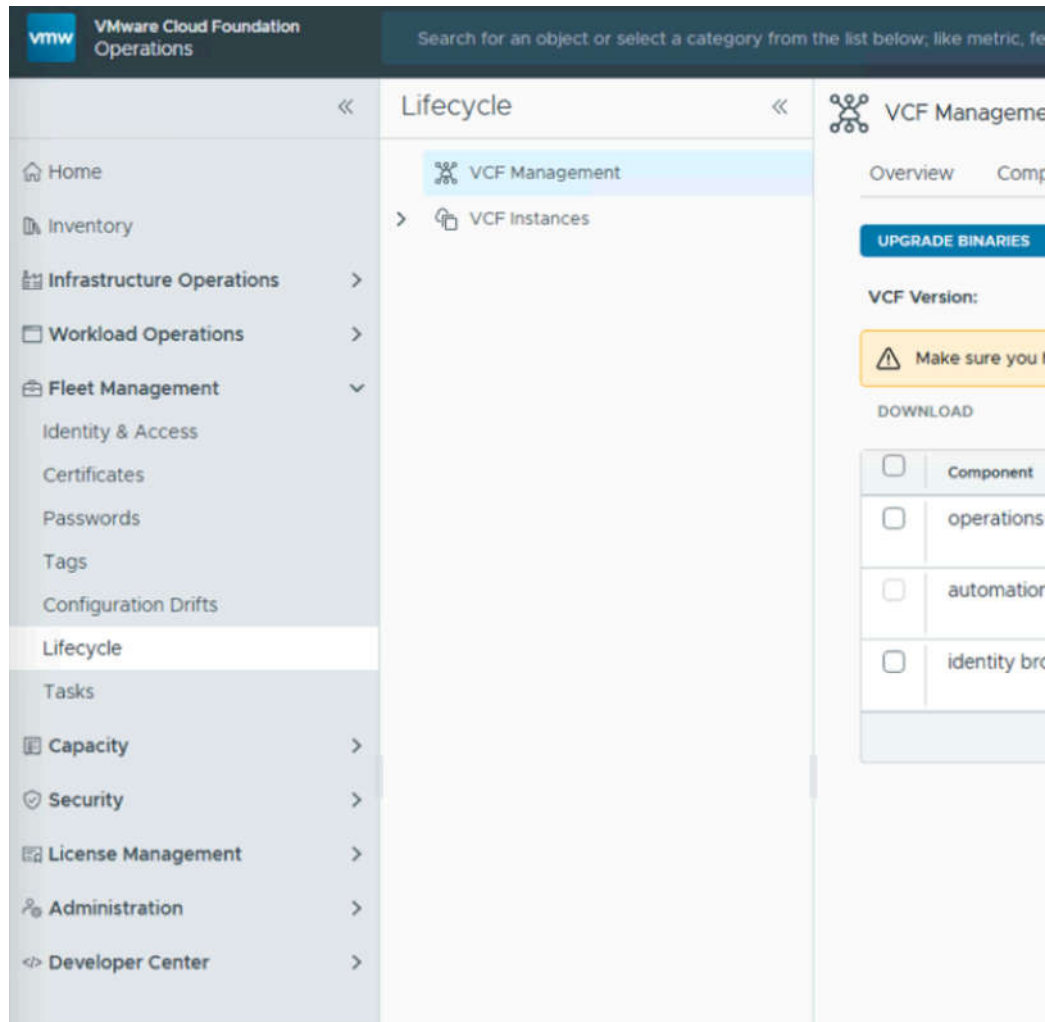
Before you download a binary, you must set up your depot configuration. See [Lifecycle Management of VCF Management Components](#).

To download a binary, select **Fleet Management > Lifecycle > Binary Management** and select the binary type, the VCF version, and the component for the request. Before you click **DOWNLOAD**, ensure that you have selected the correct upgrade, patch, or install binary type. In that way, the correct button appears when you are ready to upgrade, manage, or add your component.

To remove a binary, check the downloaded component and click the trashcan icon.

Figure 157: Binary Management tab

Select the component binary to download or remove.



Upgrade or Patch the VCF Management Components

Use VCF Operations to upgrade or patch the VCF management components.

Before upgrading or patching the VCF management components, you must generate a download token and must configure and connect to an offline or an online depot.

You then download the required binaries and upgrade or patch the VCF Operations fleet management appliance to the respective compatible version. VCF Operations fleet management appliance must be upgraded or patched before you can upgrade or patch the remaining VCF management components.

Patch VCF Operations Fleet Management

You first patch the VCF Operations fleet management appliance to the desired target version for all management components, before you can upgrade or patch them to that target version.

- Download the VCF Operations fleet management patch binaries.
 - In VCF Operations, in the left pane, navigate to **Fleet management > Lifecycle**.
 - In the **Lifecycle** pane, select **VCF Management**.
 - On the **VCF Management** page, click the **Binary management** tab.
 - Click the **Patch binaries**, select the **fleet management** component, and click **Download**.
- Create a snapshot for VCF Operations fleet management.
 - On the **VCF Management** page, click the **Settings** tab.
 - In the left pane, navigate to **System administration > System patches** and click **Create snapshot**.
 - In the **Create snapshot** dialog box, enter the vCenter hostname, select the respective credentials, and click **Submit**.
- Patch VCF Operations fleet management.
 - On the **VCF Management** page, click the **Settings** tab.
 - In the left pane, navigate to **System administration > System patches** and click **New patch**.
 - In the **Install patch** dialog box, on the **Select patch** page, select the patch and click **Next**.
 - On the **Review and install** page, click **Install**.

Plan Upgrade for the VCF Management Components

After VCF Operations fleet management is on the target version, you can download the necessary upgrade or patch binaries and plan the upgrade for the VCF management components.

- Download VCF management component binaries.
 - In VCF Operations, in the left pane, navigate to **Fleet management > Lifecycle**.
 - In the **Lifecycle** pane, select **VCF Management**.
 - On the **VCF Management** page, click the **Binary management** tab.
 - Depending on whether you are upgrading or patching the VCF Management components, click **Upgrade binaries** or **Patch binaries**.

For VCF 9.0.1, you can find the VCF Management components binaries under both patch and upgrade binaries.

Upgrade Binaries	Patch Binaries
- VCF Operations - VCF Operations for logs	- VCF Identity Broker - VCF Automation

Upgrade Binaries	Patch Binaries
• VCF Operations for networks	

- e) Select the target components and click **Download**.
2. Create a plan for the VCF management components upgrade.
 - a) On the **VCF Management** page, click the **Components** tab and click **Plan upgrade**.
 - b) In the **Plan upgrade** dialog box, from the **VCF** drop-down menu, select the VCF version, select the target version for each of the VCF management components, and click **Create plan**.

Upgrade an Individual VCF Management Component

Use VCF Operations to upgrade an individual VCF Management component that you previously included in an upgrade plan.

Before you upgrade, it is a best practice that you back up the component. The upgrade process by default creates a snapshot of the components that support it.

1. In VCF Operations, in the left pane, navigate to **Fleet management > Lifecycle**.
2. In the **Lifecycle** pane, select **VCF Management**.
3. On the **VCF Management** page, click the **Overview** tab.
4. In the VCF Management component card, click **Upgrade**.
5. Synchronize the VCF Operations inventory.
 - a) In the **Proceed to upgrade** dialog box, click **Trigger inventory sync**.
 - b) In the **Trigger inventory sync** dialog box, click **Submit**.
6. Upgrade the VCF Management component.
 - a) On the **VCF Management** page, click the **Overview** tab.
 - b) In the VCF Management component card, click **Upgrade**.
 - c) In the **Proceed to upgrade** dialog box, click **Proceed**.

The **Upgrade** wizard opens.
 - d) On the **Select version** page, click **Next**.
 - e) On the **Upgrade assessment** page, click **Run assessment**.
 - f) After the assessment completes successfully, select the **I have viewed the report and agree to proceed** checkmark and click **Next**.
 - g) On the **Snapshot** page, select the checkmarks for taking and retaining a snapshot of the component, and click **Next**.
 - h) On the **Precheck** page, click **Run precheck**.
 - i) After the precheck completes successfully, click **Next**.
 - j) On the **Upgrade summary** page, click **Submit**.

Patch an Individual VCF Management Component

Use VCF Operations to patch an individual VCF Management component that you previously included in an upgrade plan.

As a best practice, ensure that your backup and restore settings are configured properly and that your SFTP server is operational. During the patch process, VCF Operations automatically backs up the component to provide a recovery point if there is a failure.

1. In VCF Operations, in the left pane, navigate to **Fleet management > Lifecycle**.
2. In the **Lifecycle** pane, select **VCF Management**.
3. On the **VCF Management** page, click the **Overview** tab.
4. In the VCF Management component card, click **Apply patch**.
5. In the **Install patch** dialog box, on the **Select patch** page, select the patch and click **Next**.
6. On the **Review and install** page, click **Install**.

Lifecycle Management of VCF Core Components

Use VCF Operations to manage upgrade, patch, and install binaries for SDDC Manager, vCenter, NSX Manager, and ESX.

Before you can use the binaries for lifecycle management, you must download them to SDDC Manager.

You can also use VCF Operations to manage ESX components and vSphere Lifecycle Manager images.

Update Order for Core Components for Maintenance Releases

When you update your VCF environment to a maintenance release version, for example, from 9.0.0.0 to 9.0.1.0 or 9.0.2.0, you first update your management components that operate on the fleet level before you update the core components in your instances.

For the management components, start with the VCF Operations fleet management appliance, followed by the VCF Operations instance and then proceed with the remaining components in your preferred order. See [Lifecycle Management of VCF Management Components](#).

For the core components, follow the specified update order:

1. SDDC Manager
2. NSX
3. vCenter
4. ESX hosts
5. vSAN

Update or Patch SDDC Manager from VCF Operations

You can update or patch only SDDC Manager without having to upgrade the entire VCF Instance. An independent SDDC Manager patch release includes a change in the fourth group of digits in its version number, for example, SDDC Manager 9.0.0.0100.

- Download the SDDC Manager binary.

You can upgrade SDDC Manager without upgrading the VCF Instance when:

- The target version of SDDC Manager is compatible with all the BOM versions running in your VCF Instance (management and workload domains).
- There is a supported upgrade path from your current SDDC Manager version to the target SDDC Manager version.

Note: You can use the SDDC Manager upgrade functionality to upgrade SDDC Manager even when the target version of SDDC Manager is part of a full VCF BOM release, as long as it is compatible.

Updating only SDDC Manager, does not change the version of the management domain.

1. In VCF Operations, browse to **Fleet Management > Lifecycle**.

2. Select a VCF Instance and click **SDDC Manager Updates**.

The UI shows available SDDC Manager updates that are either SDDC Manager-only updates or SDDC Manager updates that are part of a full VCF Instance update.

3. Review and address any compatibility warnings.

4. Click **Run Precheck**.

Resolve any precheck issues before proceeding.

5. Schedule the update to run now or at a specific time and click **Start Update**.

Upgrade or Update the Core Components in your VCF Domain

You can use the upgrade planner in VCF Operations to select specific target versions for each VCF core component that you want to upgrade or update in a VCF domain.

If you are connected to an online or offline depot, download the binaries for the target versions of each VCF core component. See [Download VCF Core Component Binaries From an Online or Offline Depot](#).

If you are not connected to an online or offline depot, see [Download Upgrade Binaries for a Subset of Core Components Without an Online or Offline Depot](#) to download the metadata files required to plan an upgrade.

You can use the upgrade planner to select any supported version for each of the VCF core components.

1. In VCF Operations, browse to **Fleet Management > Lifecycle**.

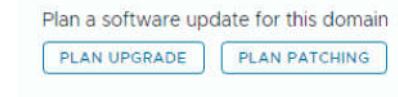
2. Expand **VCF Instances** and browse to a VCF domain in your VCF Instance.

3. Click **Updates**.

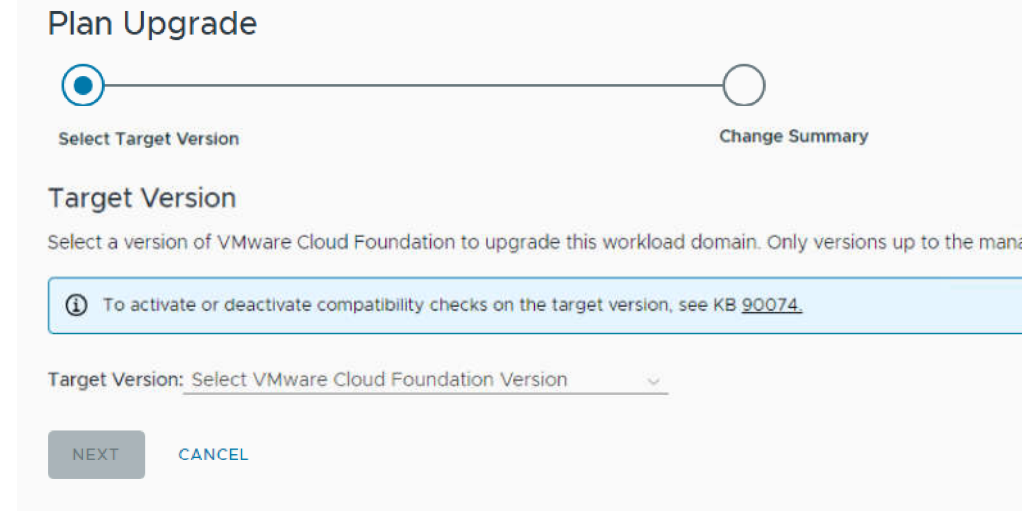
4. Click **Precheck** to run the upgrade precheck.

Resolve any issues before proceeding with an upgrade.

5. In the Available Updates section, click **Plan Upgrade** to create a new upgrade plan.



6. Select the target version of VMware Cloud Foundation from the drop-down menu and click **Next**.



7. Click **Customize Upgrade** to select specific target versions for each VCF core component.

8. Use the drop-down menus in the Target Version column to select a target version for each component and then click **Validate Selection**.

Change Summary

The following product upgrades will be planned out. The target versions of individual products can be customized if necessary.

VALIDATE SELECTION		RESET TO DEFAULTS		
Software Component	Current Version	Current Build	Target Version	Target Build
VMware ESXi	7.0.3	20326353	8.0.3-24017672 --	24017672
VMware NSX	3.2.1.2.0	20541212	4.2.0.0.0-24013905 --	24013905
VMware vCenter Server Appliance	7.0.3.01000	20395099	8.0.3.00000-24017671 --	24017671

CONFIRM BACK CANCEL

9. After validation succeeds, click **Confirm**.

10. Review the update sequence based on your target version selections and click **Done**.

Note: If you do not have an online or offline depot, you need to use the VCF Download Tool to download the binaries for the VCF BOM component target versions before you can update them. See [Download Upgrade Binaries for a Subset of Core Components Without an Online or Offline Depot](#).

11. In the Available Updates screen, click **Schedule Update** or **Update Now** to update the first component.

Continue to update the VCF core components until they are all updated.

Updating or Patching Individual VCF Core Components

You can use the patch planner functionality in VCF Operations for updating or patching individual VCF core components in a VCF domain.

Download the patch binaries. The method you use to download binaries depends on how you access the Internet in your environment.

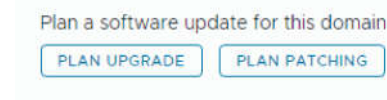
- [Download VCF Core Component Binaries to an Offline Depot](#)
- [Download VCF Core Component Binaries From an Online or Offline Depot](#)
- [Download Upgrade Binaries for All Core Components Without an Online or Offline Depot](#)

The patch planner provides the ability to apply update or patch binaries to VCF core components. If you are connected to the online depot, binaries are available in the patch planner. If you do not have access to an online depot, use the VCF Download Tool to download binaries. See [Download VCF Core Component Binaries to an Offline Depot](#) or [Download Upgrade Binaries for All Core Components Without an Online or Offline Depot](#).

1. In VCF Operations, browse to **Fleet Management > Lifecycle**.
2. Expand **VCF Instances** and browse to a workload domain in your VCF instance.
3. Click **Updates**.
4. Click **Precheck** to run the upgrade precheck.

Resolve any issues before proceeding with an upgrade.

5. In the Available Updates section, click **Plan Patching** create a new plan for the desired components.



Note:

You cannot use the patch planner if you have an existing upgrade plan. Cancel the upgrade plan to create a patching plan.

6. Select the components to patch and the target versions and then click **Validate Selection**.

Note: When you select a target vCenter version, the UI indicates which versions support vCenter Reduced Downtime Upgrade (RDU).

7. After validation succeeds, click **Confirm**.

8. Review the update sequence based on your target version selections and click **Done**.

9. In the Available Updates screen, click **Schedule Update** or **Update Now** to update the first component.

Continue to update the remaining planned components until they are all updated.

Managing vSphere Lifecycle Manager Images for VMware Cloud Foundation

VMware Cloud Foundation uses vSphere Lifecycle Manager images to apply software and firmware updates to the ESX hosts in an SDDC cluster.

You create vSphere Lifecycle Manager images in the vSphere Client and then import them into a VCF Instance using VCF Operations. You can use imported images when you:

- Upgrade the management domain or a workload domain.
- Create a workload domain.
- Add an SDDC cluster to the management domain or a workload domain.

You can create a composite image for a cluster that has hosts from different vendors or from the same vendor but different generations, family, or model. A composite image contains one default image and one or more alternative images. All images in a composite image contain the same base image and solutions, but may include different vendor or firmware and driver add-ons. When you create alternate images, you also specify the rules that are used to filter the hosts to which the alternative image is assigned.

For more information about vSphere Lifecycle Manager images, see [Managing Host and Cluster Lifecycle](#) in the vSphere documentation.

If you are using an offline depot or no depot, you must use the VCF Download Tool to ensure that the latest ESX component data is available for creating vSphere Lifecycle Manager images in the vSphere Client.

- [Download ESX Component Data to an Offline Depot](#)
- [Download ESX Component Data Without an Online or Offline Depot](#)
- [Synchronize ESX Component Data for a VCF Instance](#)

Import a vSphere Lifecycle Manager Image to a VCF Instance

After you create vSphere Lifecycle Manager images in the vSphere Client, you can import those images into VCF Operations and use them for creating and updating VCF domains and clusters.

Create a vSphere Lifecycle Manager image using the vSphere Client. See [Working with Image Library](#) in the vSphere documentation.

- Use the ESX version specified in the VMware Cloud Foundation BOM.
- You can create a composite image for a cluster that has hosts from different vendors or from the same vendor but different generations, family, or model. A composite image contains one default image and one or more alternative images. All images in a composite image contain the same base image and solutions, but may include different vendor or firmware and driver add-ons. When you create alternate images, you also specify the rules that are used to filter the hosts to which the alternative image is assigned.
- If you plan to use the **Import from a file** option to import a vSphere Lifecycle Manager image, you must first export the files from the vSphere Client. See [Working with Image Library](#) in the vSphere documentation.

1. In VCF Operations, click **Fleet Management > Lifecycle**.
2. Click **VCF Instances** and click a VCF Instance name.
3. Click **Image Management**.
4. Click **Import Image**.
5. Select the import method, provide the required information, and click **Import**.

Option	Description
Import from a vCenter	Select the vCenter where you created the image from the drop-down menu. Select the image to import.
Import from a file	Enter a name for the image and then click Select File to import each of the required files. Select the JSON, ZIP, and ISO files from your local computer. The ISO file is optional.

Download ESX Component Data to an Offline Depot

Use the VCF Download Tool Update Manager Download Service (UMDS) commands to download ESX component data to the offline depot which any VCF Instance connected to the offline depot can access.

- [Set Up an Offline Depot Web Server for VMware Cloud Foundation](#).
- Generate a download token on the Broadcom Support Portal.
- The offline depot must have:
 - The latest version of the VCF Download Tool. You can download it from the Broadcom Support portal.
 - Internet connectivity (either directly or through a proxy) for downloading the bundles and other required files.
 - Configure TCP keepalive in your SSH client to prevent socket connection timeouts when using the VCF Download Tool for long-running operations.
 - If you are using a Windows computer as your offline depot, you must install Windows Subsystem for Linux (WSL) and run the UMDS commands from within WSL.
- Connect to the offline depot. See [Connect to an Offline Depot for a VCF Instance](#).

Note: You can also connect to the offline depot after you download ESX component data to the offline depot.

SDDC Manager 9.0 acts as an Update Manager Download Service (UMDS) repository in vSphere Lifecycle Manager and it is automatically added as a download source for ESX component data in vSphere. In order to have access to the most current ESX component data to create vSphere Lifecycle Manager images in the vSphere Client, you must download ESX component data to the offline depot.

If the offline depot can only access the Internet using a proxy server, use the proxy server options as described in [VCF Download Tool Update Manager Download Service \(UMDS\) Commands](#).

Note: VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

1. On the computer hosting offline depot, run the following command to install UMDS.

```
./vcf-download-tool umds install
```

Note: You can skip this step if UMDS is already installed.

2. Configure the patch store for the offline depot.

```
./vcf-download-tool umds run -S --patch-store offline_depot_root_dir/umds-patch-store/
```

Note:

Replace *offline_depot_root_dir* with the root directory of your offline depot. For example: `./vcf-download-tool umds run -S --patch-store /var/www/umds-patch-store/`.

3. Configure the download token.

```
./vcf-download-tool umds run -S --add-entitlement-token
```

4. Enable downloading of ESX host updates.

```
./vcf-download-tool umds run -S --enable-host
```

5. (Optional) Configure UMDS to connect to the websites of third-party vendors to download vendor-specific ESX host patches and notifications.

```
./vcf-download-tool umds run -S --add-url https://host_url/index.xml --url-type HOST
```

6. Download updates based on the current configuration.

```
./vcf-download-tool umds run vmware-umds -D
```

7. (Optional) Import third-party offline zip files.

```
./vcf-download-tool umds run --import --offline-bundle absolute_path_to_zip_file
```

After the ESX component data is available in the offline depot, you can use the vSphere Client to create a vSphere Lifecycle Manager image and import that image to a VCF Instance.

Download ESX Component Data Without an Online or Offline Depot

If you do not have access to an online or offline depot, you can use the VCF Download Tool Update Manager Download Service (UMDS) commands to download ESX component data to a different computer and then upload the data to the SDDC Manager appliance.

- Generate a download token on the Broadcom Support Portal.
- A Windows or Linux computer with Internet connectivity (either directly or through a proxy) for downloading the bundles and other required files.
 - Note:** If you are using a Windows computer, you must install Windows Subsystem for Linux (WSL) and run the UMDS commands from within WSL.
- The latest version of the VCF Download Tool available on both the computer with Internet access and the SDDC Manager appliance. You can download the VCF Download Tool from the Broadcom Support Portal.
- Configure TCP keepalive in your SSH client to prevent socket connection timeouts for long-running operations.

Note: The VCF Download Tool is the only supported method for downloading bundles. Do not use third-party tools or other methods to download bundles.

SDDC Manager 9.0 acts as an Update Manager Download Service (UMDS) repository in vSphere Lifecycle Manager and it is automatically added as a download source for ESX component data in vSphere. In order to have access to the most current ESX component data to create vSphere Lifecycle Manager images in the vSphere Client, you must download ESX component data to a computer with Internet access and upload that data on the SDDC Manager appliance.

If the computer with Internet access can only access the Internet using a proxy server, use the proxy server options as described in [VCF Download Tool Update Manager Download Service \(UMDS\) Commands](#).

Note: VMware Cloud Foundation does not support proxy servers with NTLMv2 authentication.

1. On the computer with Internet access, run the following command to install UMDS.

```
./vcf-download-tool umds install
```

Note: You can skip this step if UMDS is already installed.

2. Enable downloading of ESX host updates.

```
./vcf-download-tool umds run -S --enable-host
```

3. (Optional) Configure UMDS to connect to the websites of third-party vendors to download vendor-specific ESX host patches and notifications.

```
./vcf-download-tool umds run -S --add-url https://host_url/index.xml --url-type HOST
```

4. Download updates based on the current configuration.

```
./vcf-download-tool umds run vmware-umds -D
```

5. Export the downloaded data.

```
./vcf-download-tool umds run -E --export-store repository_path
```

The *repository_path* can be the path to a portable media drive. Export the downloads to the portable media drive that you can use to physically transfer the patches to the SDDC Manager appliance.

6. Copy the exported data to a location on the SDDC Manager appliance.

7. SSH in to the SDDC Manager appliance using the `vcf` user account.

8. Navigate to `/opt/vmware/vmware-umds/bin`.

9. Import the data.

```
vmware-umds --import --import-store repository_path
```

Replace *repository_path* with the location to which you copied the data.

10. (Optional) Import a single third-party add-on or component as an offline zip file.

```
vmware-umds -I --offline-bundle absolute_path_to_zip_file
```

Replace *absolute_path_to_zip_file* with the path to the zip file. For example: `/tmp/VMware-ESXi-9.0.0-24197573-vib-test-certs.zip`.

Synchronize ESX Component Data for a VCF Instance

vSphere Lifecycle Manager synchronizes to its download sources, including SDDC Manager, regularly and automatically, but you can also trigger synchronization at any time from VCF Operations.

To ensure that the most recent ESX component data is available for creating vSphere Lifecycle Manager images, you can update the contents of the vSphere Lifecycle Manager depot by synchronizing it with the SDDC Manager in a VCF Instance.

1. In VCF Operations, click **Fleet Management** > **Lifecycle**.
2. Click **VCF Instances** and click on the name of a VCF Instance.
3. Click **Binary Management**.
4. In the **ESX Components** section, click **Synchronize Now**.

Upgrading VMware Cloud Foundation 5.x Workload Domains to 9.0

To upgrade to a VMware Cloud Foundation 5.x workload domain to 9.0, all workload domains must be at VMware Cloud Foundation 5.0 or higher. If any workload domain is at a version lower than 5.0, you must upgrade it to 5.0 and then upgrade to 9.0.

You can use VCF Operations or the SDDC Manager UI to upgrade your VMware Cloud Foundation 5.x workload domains to 9.0. The following procedures use the SDDC Manager UI.

Note: The SDDC Manager UI is being deprecated and will be removed in a future release. After upgrading your workload domains to 9.0, use VCF Operations to perform lifecycle management of VMware Cloud Foundation components.

The steps and the order of steps to upgrade a workload domain are similar to the steps to upgrade the management domain.

Task	Procedure
1. Plan the upgrade.	Plan a Workload Domain Upgrade
2. Perform an update precheck.	Perform Update Precheck in SDDC Manager
3. Upgrade NSX in a federated environment.	Upgrading NSX in a Federated Environment
4. Upgrade NSX.	Upgrading NSX to 9.0
5. Upgrade vCenter.	Upgrade to vCenter 9.0 for VMware Cloud Foundation
6. Transition from vLCM baselines to vLCM images.	Transitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images
7. Upgrade ESX.	Upgrade to ESX 9.0 for VMware Cloud Foundation
8. Upgrade vSphere Supervisor on clusters that have Workload Management enabled.	See "Updating vSphere Supervisor" in the vSphere documentation.
9. Upgrade vSphere Distributed Switch version	Upgrade vSphere Distributed Switch Versions
10. Upgrade vSAN on-disk format versions.	Update vSAN On-Disk Format Versions
11. Post upgrade steps for NFS-based workload domains	Post Upgrade Steps for NFS-Based Workload Domains

After all upgrades have completed successfully:

1. Remove the VM snapshots you took before starting the update.
2. Take a backup of the newly installed components.

Plan a Workload Domain Upgrade

Before proceeding with a workload domain upgrade you must first plan the upgrade to your target version in the SDDC Manager UI.

Upgrade the management domain to VMware Cloud Foundation 9.0. See [Upgrade the Management Domain Core Components to VMware Cloud Foundation 9.0](#).

1. In the navigation pane, click **Inventory** > **Workload Domains**.
2. On the **Workload Domains** page, click the workload domain you want to upgrade and click the **Updates** tab.
3. Under **Available Updates**, click **PLAN UPGRADE**.
4. On the **Plan Upgrade for VMware Cloud Foundation** screen, select the target version from the drop-down, and click **NEXT**.

Bundles applicable to the chosen release will be made available to the workload domain.

Post Upgrade Steps for NFS-Based Workload Domains

After upgrading workload domains that use NFS storage, you must add a static route for hosts to access NFS storage over the NFS gateway. This process must be completed before expanding the workload domain.

1. Identify the IP address of the NFS server for the workload domain.
2. Identify the network pool associated with the hosts in the cluster and the NFS gateway for the network pool.
 - a) Log in to SDDC Manager.
 - b) Click **Inventory > Workload Domains** and then click the workload domain.
 - c) Click the **Clusters** tab and then click an NFS-based cluster.
 - d) Click the **Hosts** tab and note down the network pool for the hosts.
 - e) Click the Info icon next to the network pool name and note down the NFS gateway.
3. Ensure that the NFS server is reachable from the NFS gateway. If a gateway does not exist, create it.
4. Identify the vmknics on each host in the cluster that is configured for NFS traffic.
5. Configure a static route on each host to reach the NFS server from the NFS gateway.

```
esxcli network ip route ipv4 add -g NFS-gateway-IP -n NFS-gateway
```

6. Verify that the new route is added to the host using the NFS vmknics.

```
esxcli network ip route ipv4 list
```

7. Ensure that the hosts in the NFS cluster can reach the NFS gateway through the NFS vmkernel.

For example:

```
vmkping -4 -I vmk2 -s 1470 -d -W 5 10.0.22.250
```

8. Repeat steps from 2 through 7 for each cluster using NFS storage.